

PCT/PCM6: Three Point Pairs

4. Registration Before/After

The alignment worked because the difference image after registration contains much less visible structure than the difference image before registration. Most scene details cancel when the two images are correctly aligned. A small rotation error would leave double edges or thin, opposite-side residuals around objects, especially near high-contrast boundaries.

5. Minimum Correspondence Pairs

For an affine transform, there are 6 unknowns and each point pair provides 2 equations:
 $6 / 2 = 3$ pairs.

Therefore, the minimum is 3 point pairs.

For a homography, there are 8 unknowns:
 $8 / 2 = 4$ pairs.

Therefore, the minimum is 4 point pairs.

Practitioners use many more than the minimum because real correspondences contain measurement errors and outliers. Extra points provide redundancy and make the estimated transformation more reliable.

6. Prediction

If one clicked pair is wrong by 30 pixels, ordinary least-squares fitting will be pulled toward the erroneous correspondence rather than staying exactly near the true transformation. This happens because squared error gives large residuals disproportionately large influence: a residual of 30 contributes $30^2 = 900$ to the objective. Therefore, even one large outlier can strongly affect the fitted transform.